CAAD: Education-Research and Practice
ECAADE Conference 1989
School of Architecture in Aarhus Denmark

COMPUTER GRAPHICS IN ARCHITECTURAL EDUCATION
THE STRATHCLYDE EXPERIENCE

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Abstract:

The following paper describes the teaching and research activities of the Department of Architecture and Building Science at the University of Strathclyde and highlights the role of CAAD and, more specifically, of Computer Graphics in the curriculum. The presentation during the Conference will deal in greater depth with how students use increasingly sophisticated hardware and software in their investigation of the formal qualities of buildings and of how best they can be visualised on site.

Background

The Department of Architecture and Building Science at the University of Strathclyde is one of 36 Schools in the UK and one of two in Glasgow, Europe's 1990 City of Culture. A number of degrees are awarded in

- Architecture (BSc, BArch)
- Building Design Engineering (BEng)
- Postgraduate Studies (MSc)
- Postgraduate Research (MPhil, PhD)

The undergraduate courses in architecture, which have about 180 students, are structured as follows:

Foundation Studies:	Year 1	
	Year 2	
Graduation Studies:	Year 3	
	Year 4	BSc
Professional Studies:	Year 5	(spent in practice)
	Year 6	Barch

All students in Foundation Studies follow the same curriculum:

Architecture Design Projects	8 credits
Building Technology & Environment	4 credits
Architecture Society & Environment	2 credits
Architectural History & Theory	2 credits
Design Theory	2 credits
Art Workshop/Visual Modelling	2 credits
Electives	4 credits

In Graduation Studies, students have a free choice from a wide range of optional subjects which fall into six broad categories, as follows:

Al	Energy Management 1
A3	Energy Management 2
B3	Contrary Planning
B4	Building Conservation
B6	Urban Design
B7	History of Modern Planning
B8	Housing - Inner City Design
B9	Form, Image and Spatial Organisation
C2	CAAD: Current Practice
C3	CAAD: Modelling Building Performance
C4	CAAD: Advanced Graphics
C5	CAAD: The Technical Bits

D1 Structures
D3 Building Fabric Performance
D4 Building Services Systems
D5 Space Planning & Interior Architecture

E1 Experimental Aesthetics
E2 Spatial Order and Social Order
E4 Scottish Architecture - Context and Content
E5 Environmental Psychology for Architects
E7 Universal Design Principles
E8 Towards an Idea - putting Humpty together

F4 The Graphic Generation of Architectural Form
F7 The Representation of Form through Freehand Drawing

Professional Studies consist of a year in practice followed by a year working individually or in groups on a single project usually set in an urban context.

What distinguishes the Strathclyde School from most of the other UK Schools is the level of research and postgraduate course work which takes place. This activity is focused on five specialist Units:

ABACUS: Architecture and Building Aids Computer Unit, Strathclyde
ESRU : Energy Simulation Research Unit
BPSU : Building Performance Studies Unit
HRRU : Housing Rehabilitation Research Unit
UDSU : Urban Design Studies Unit

The total value of the research portfolio is around £1.5 million e.c.u., with most of the contracts held by ABACUS and ESRU.

Full- and part-time MSc courses are offered by three of the Units: Computer Aided Building Design (ABACUS), Facilities Management (BPSU) and Urban Design (UDSU). The total postgraduate population (MSc, MPhil and PhD) is around 80.

Facilities

The Department has the usual range of facilities which are found in any School of Architecture but it also has very considerable computing facilities. These are grouped into three laboratories:

SUNLab
- 20 SUN 380 with 19" colour monitor
- 1 SUN 386i with 19" colour monitor

PCLab
- 8 IBM PS/2 Model 50 with VGA
- 6 Apple Mackintosh Plus

VAXLab
- 12 raster graphic monochrome terminals connected to VAX 380 cluster

Additionally, within ABACUS, there are specialist computing facilities for frame-grabbing and image mixing and for animation.

The Department is also well equipped with video facilities including a full U-matic editing suite, video projector and 9-monitor video wall. An area of expansion is the interfacing of video and computer equipment.

CAAD Courses

In Year 1 of Foundation Studies students are introduced, through a series of 1/2-day projects, to a range of computing utilities - word processing, spread sheets, database management systems and to the evolving CALinCAD software being developed under the EC COMETT initiative. In Year 2 the 1/2-day projects focus on design and drafting applications and students learn to use AUTOM and the range ABACUS packages for design appraisal and visualisation. Additionally, within particular subjects, students are introduced to applications programms in specific domains, e.g. energy consumption, daylight levels, etc. Access to computing facilities in Foundation Studies is mainly via the PCLab and the VAXLab.

In Graduation Studies (Years 3 and 4) students are free to elect to take some or all of the following CAAD classes:

CAAD2: Current Practice
The purpose of this course is to take stock of the hardware and software available to practices and to look in detail at how practices currently use computers, particularly in respect to office management and drafting. The lecture content is:

- Overview of CAAD
- A Profile of office Practice
- Computer Aided Draughting: a Review of Available Systems
- Comparison and Evaluation of Draughting Systems
- Anatomy of a Draughting System: 1
- Anatomy of a Draughting System: 2
- Anatomy of a Draughting System: 3
- Computer Aided Management: an Overview
- Job and Timesheet Accounting
- Specifications and Bills of Quantities
- Wordprocessing and Spreadsheets
- Desktop Publishing

CAAD3: Modelling Building Performance

This course looks at design methodology and explores the prospects and problems associated with the development of an integrated and comprehensive computer-based aid which helps architects improve the functional and formal performance of the built environment. The lecture content is:

- Models of the Design Decision making Activity
- Specification and Evaluation of Building Performance
- Data Structures for Representing Geometry 1
- Date Structures for Representing Geometry 2
- Parameters of Spatial Performance
- Parameters of Environmental Performance: Thermal
- Parameters of Environmental Performance: Light and Sound
- Integrated Software Environments
- Multi-Variate Evaluation
- Integrated Knowledge Based Systems
- Expert Systems
- Social Impacts of CAAD

CAAD4: Advanced Graphics

The course introduces an increasingly powerful set of tools for generating and viewing the natural and man-made environment. The central themes include the creation and display of building geometry, representation of terrain and vegetation, the merging of computer generated images with video images and the issue of realism in computer graphics. The lecture content is:

- Overview of Computer Graphics
- Creating Simple Geometries
- Creating Complex Geometries
- Viewing in Wireline and in Stereo
- Photomontage
- Creating a Viewpath of Images
- Rendered and Shadowed Images
- Modelling Terrain
- Modelling Vegetation
- Modelling Interiors
- Animation
- Video Mixing
-

CAAD5: The Technical Bits

This course aims to provide the student with some degree of insight into the technical issues which are central to the current capabilities and likely future developments of the hardware and software systems used in CAAD. The lecture content is:

- Anatomy of a CAAD System
- Hardware: System Architectures
- Hardware: Processors and Peripherals
- Software: Operating Systems
- Software: Languages 1
- Software: Languages 2
- Software: Tools 1
- Software: Tools 2
- Software: Engineering 1
- Software: Engineering 2
- Software: Object Oriented Programming
- Future: Innovative Hardware and Software Environments

Increasingly students in Graduation Studies work in the SUNLab environment.

The l-year MSc course in Computer Aided Building Design, which is unique in Europe, recruits between 15 and 20 students annually. The course content includes:

- Computational methods in Building Design
- Computer Applications in Integrated Design Appraisal
- Computer Applications in Dynamic Energy Modelling
- Computer Applications in Visual Modelling
- Professional Applications of Computing
- Design Methodology and Expert Systems
- Introduction to Computing
- Software Engineering
- Computer Graphics Programming for Design
- Computer Graphics in Design

MSc students predominantly use the SUNLab facilities.

Graphics Project Work

Graduation Studies students who have elected to study CAAD4: Advanced Graphics have to undertake prescribed class work in which the range of ABACUS visualisation programs (Figure A) are used to create, display and animate a simple 3-D geometry, such as a logo. The program VIM allows the geometry to be created and VIEWER produces mono or stereo perspective views (Figure B). A view path can be set up using VWALK (Figure C) and a sequence of display files created (Figure D). Using VISTA, students can shadow (Figure E) and render (Figure F) their geometry. Exceptionally, students progress to the use of DIM for the production of interior views

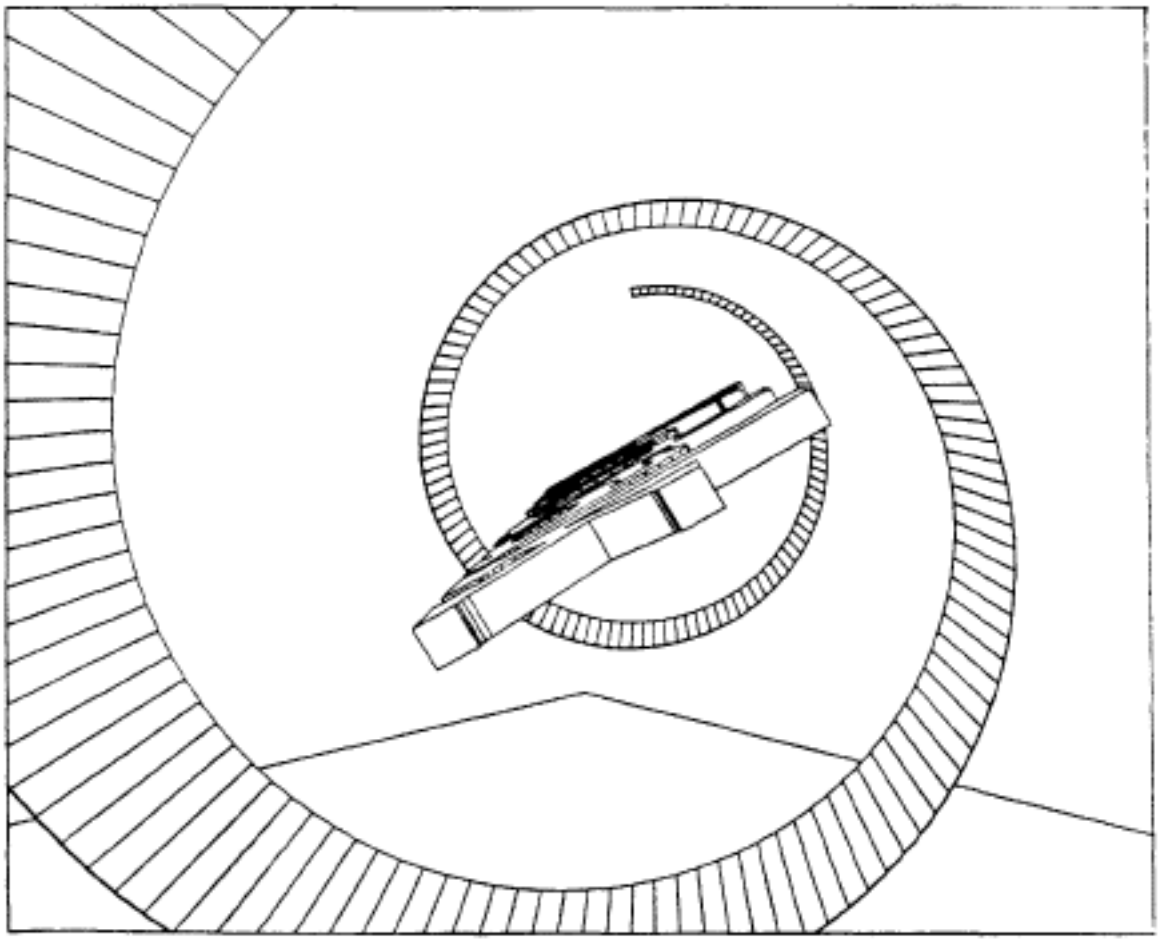
Students who successfully complete the class work can elect to undertake a 5-week project during which they have to apply the same programs in a more innovative way to the visualisation of buildings. Figures C-x-N show examples of student project work.

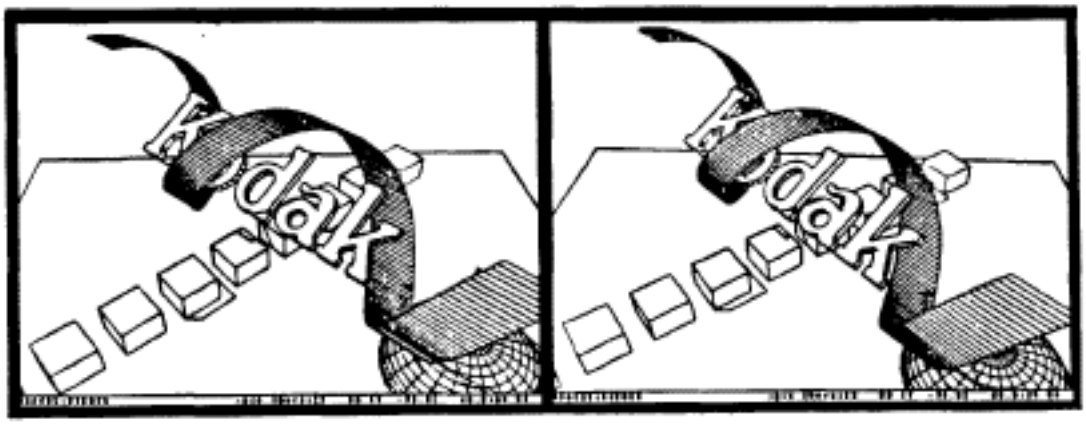
Students on the MSc (CAAD) course have concluded the formally taught part of the 1-year course by Easter. From then until the end of August they undertake a major dissertation on a project of their own choosing. Quite frequently, project work has a high graphics content, and not infrequently real innovation is achieved in the use of computer graphics in design.

The presentation during the Conference will focus on the more innovative work being achieved by undergraduate and postgraduate students.

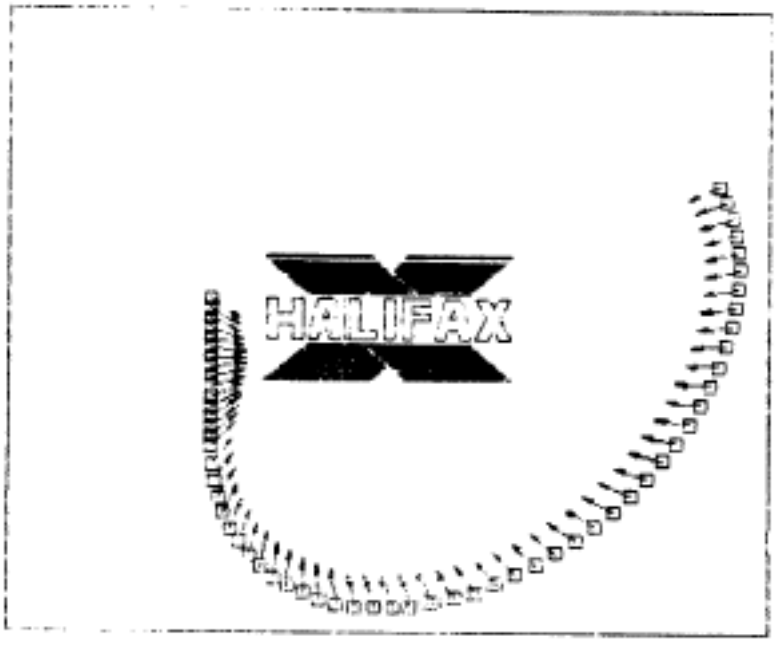
VIM
3-D Data file
VIEWER
Wireline Perspective
view
parameters
VISTA
Rendered Perspective
sun position
colours
DIM
Interior Perspective
lights
VWALK
Video
view path

A

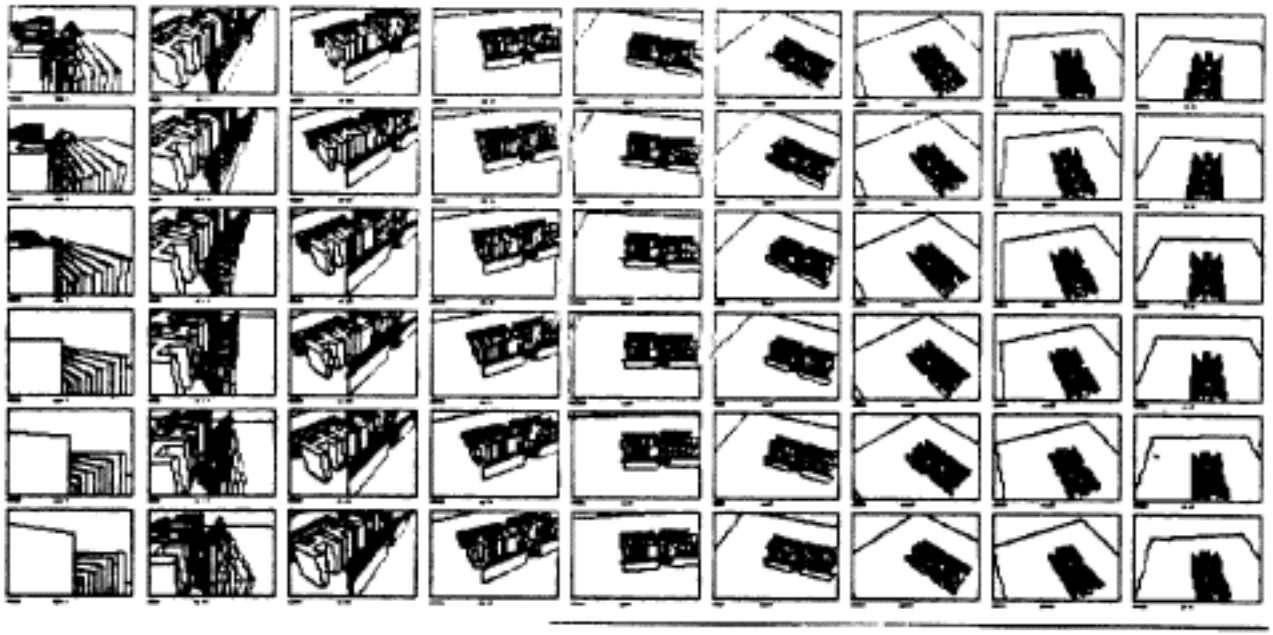




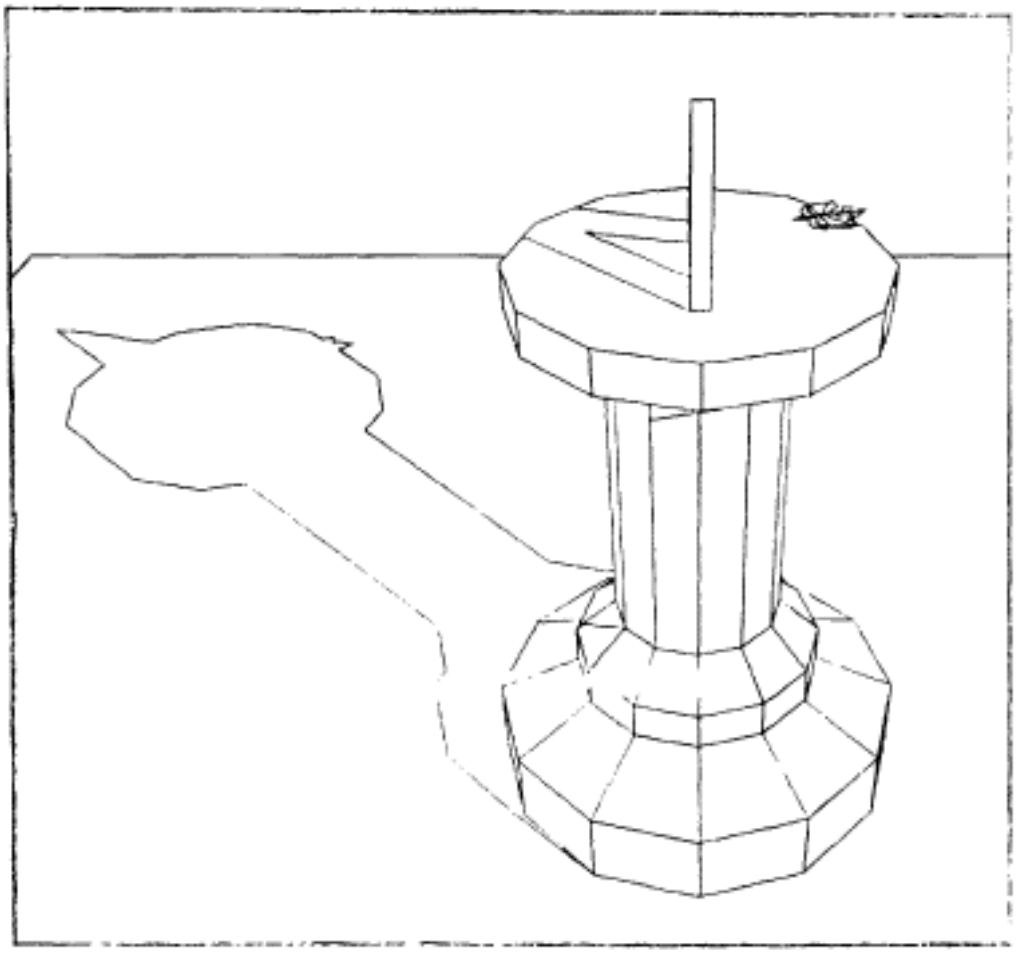
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C



D



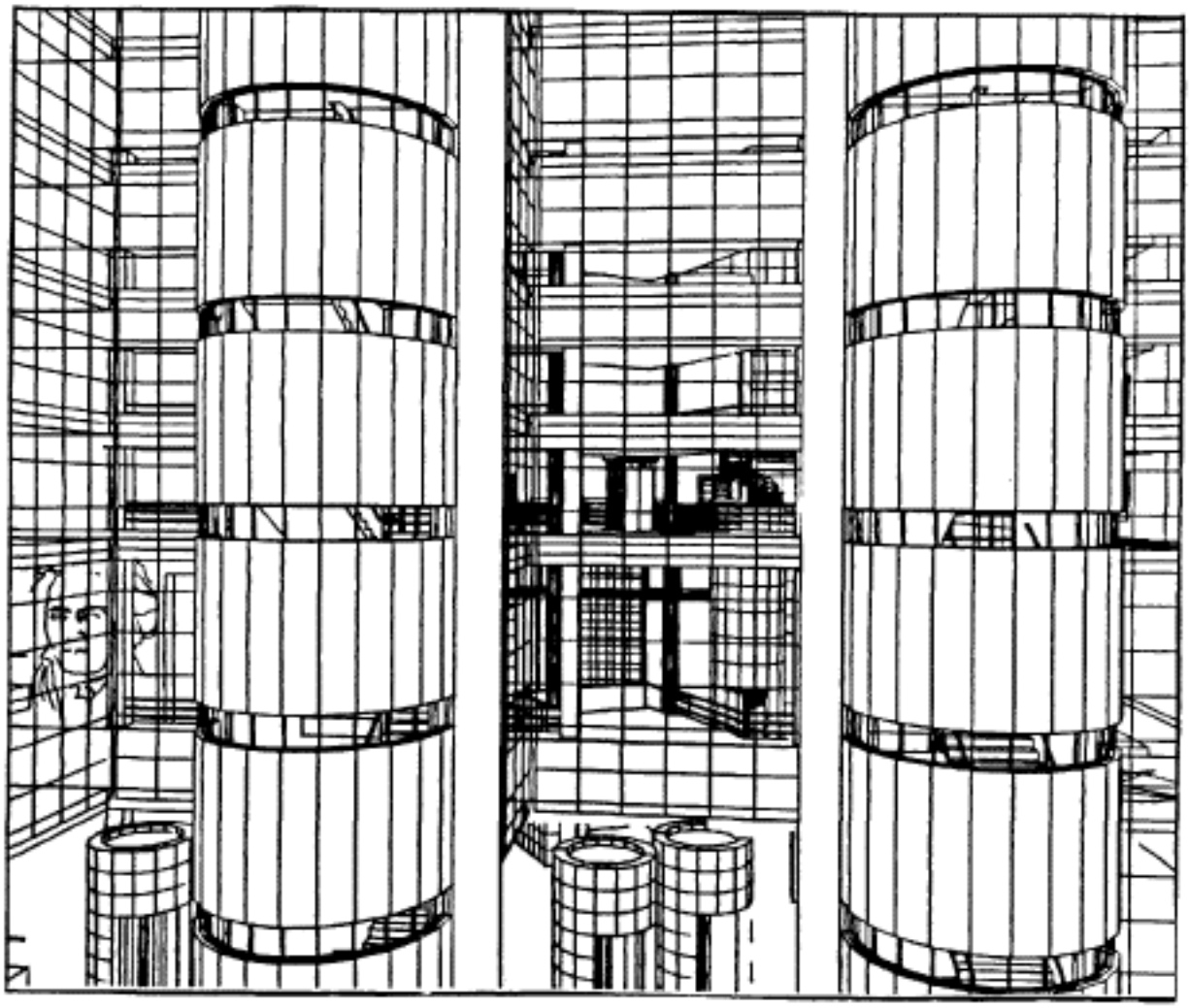
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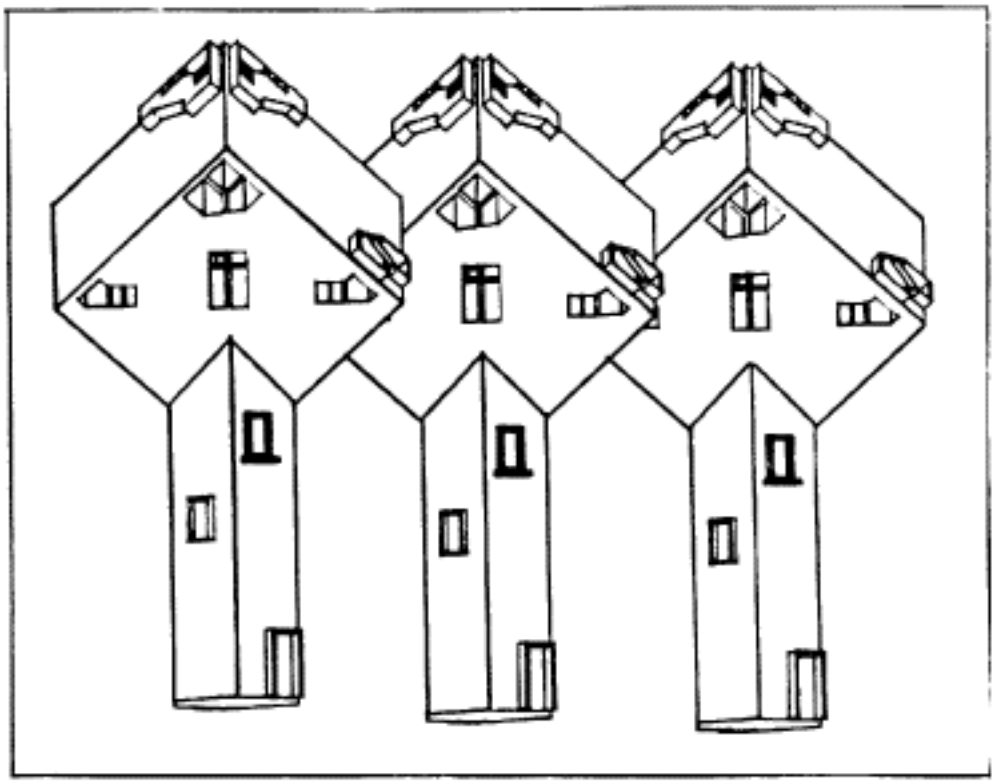
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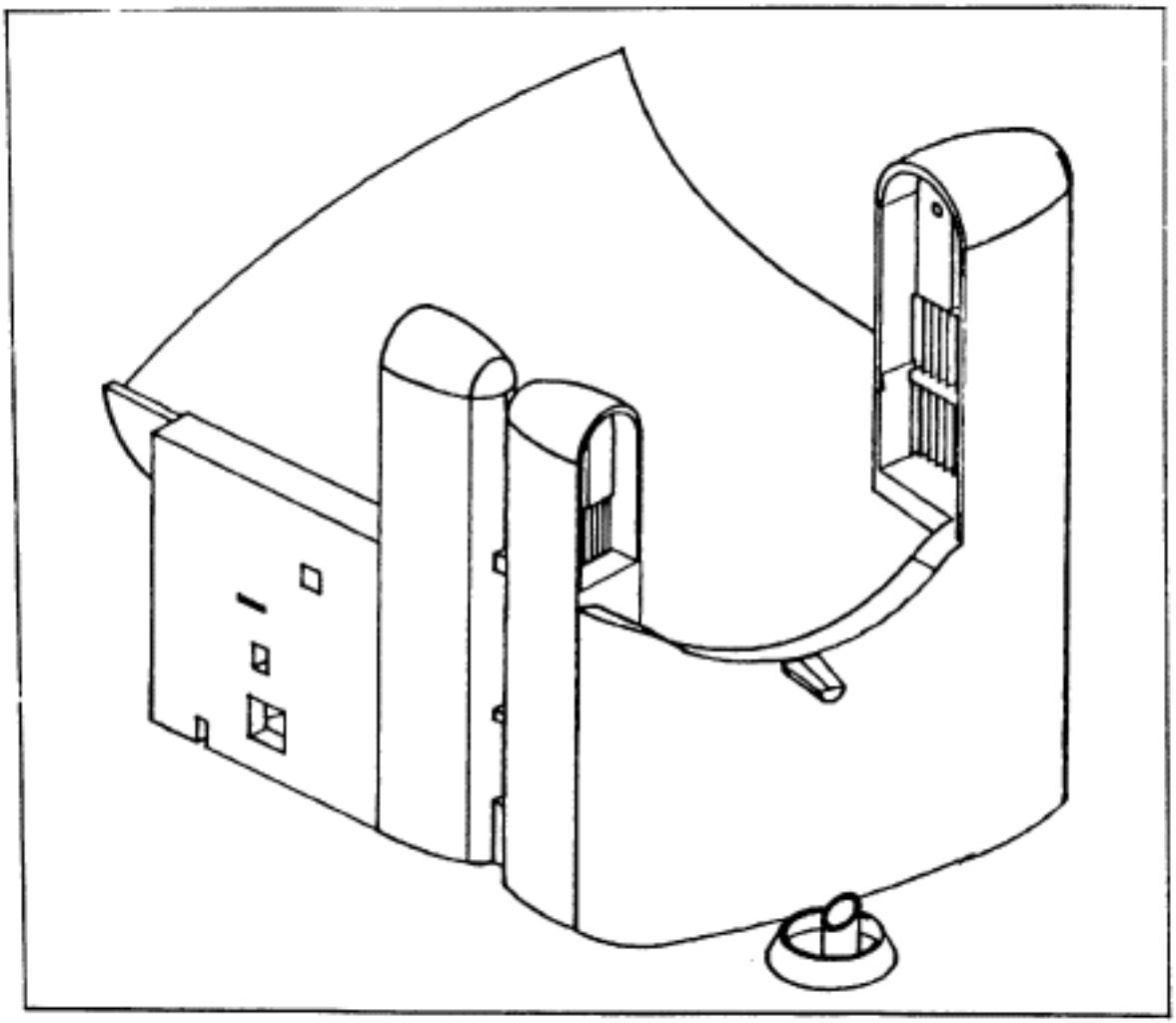


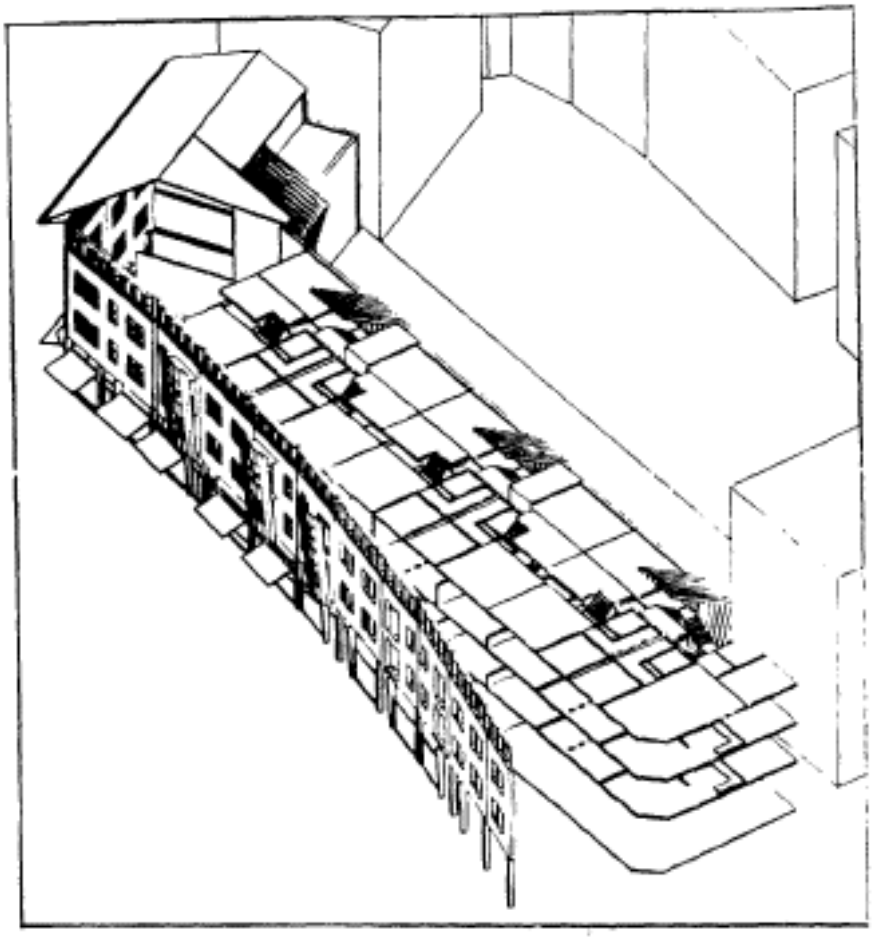
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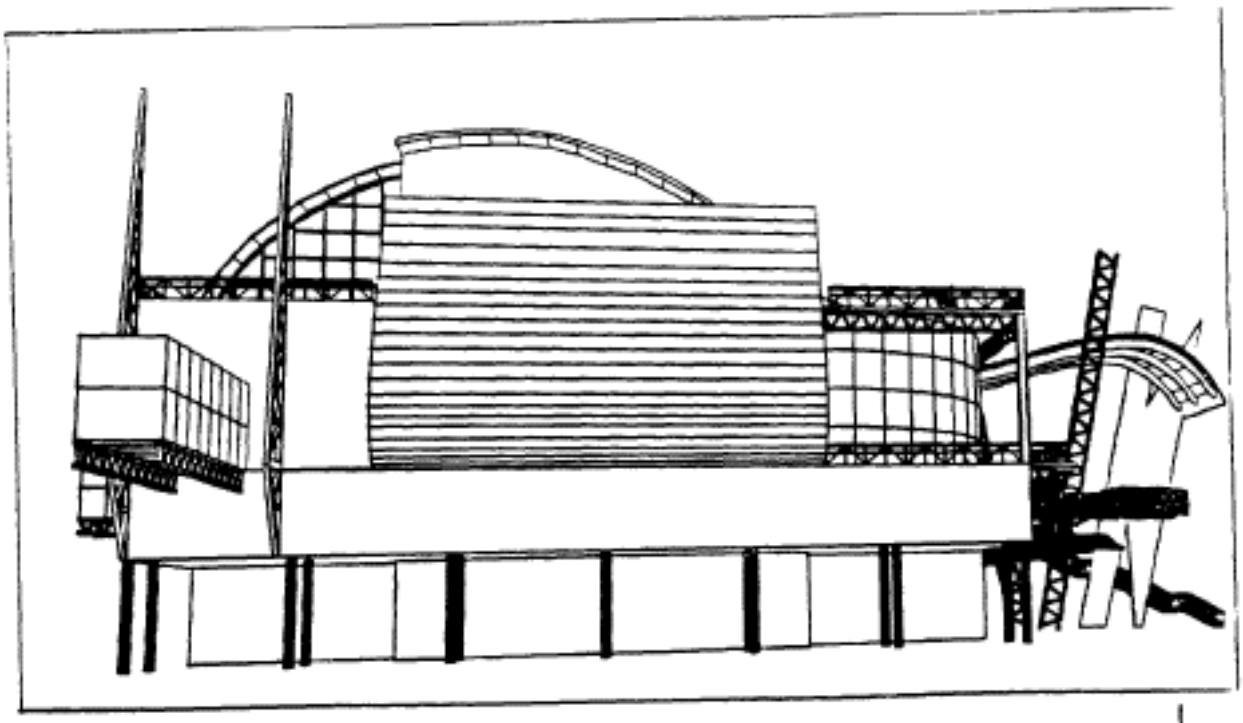
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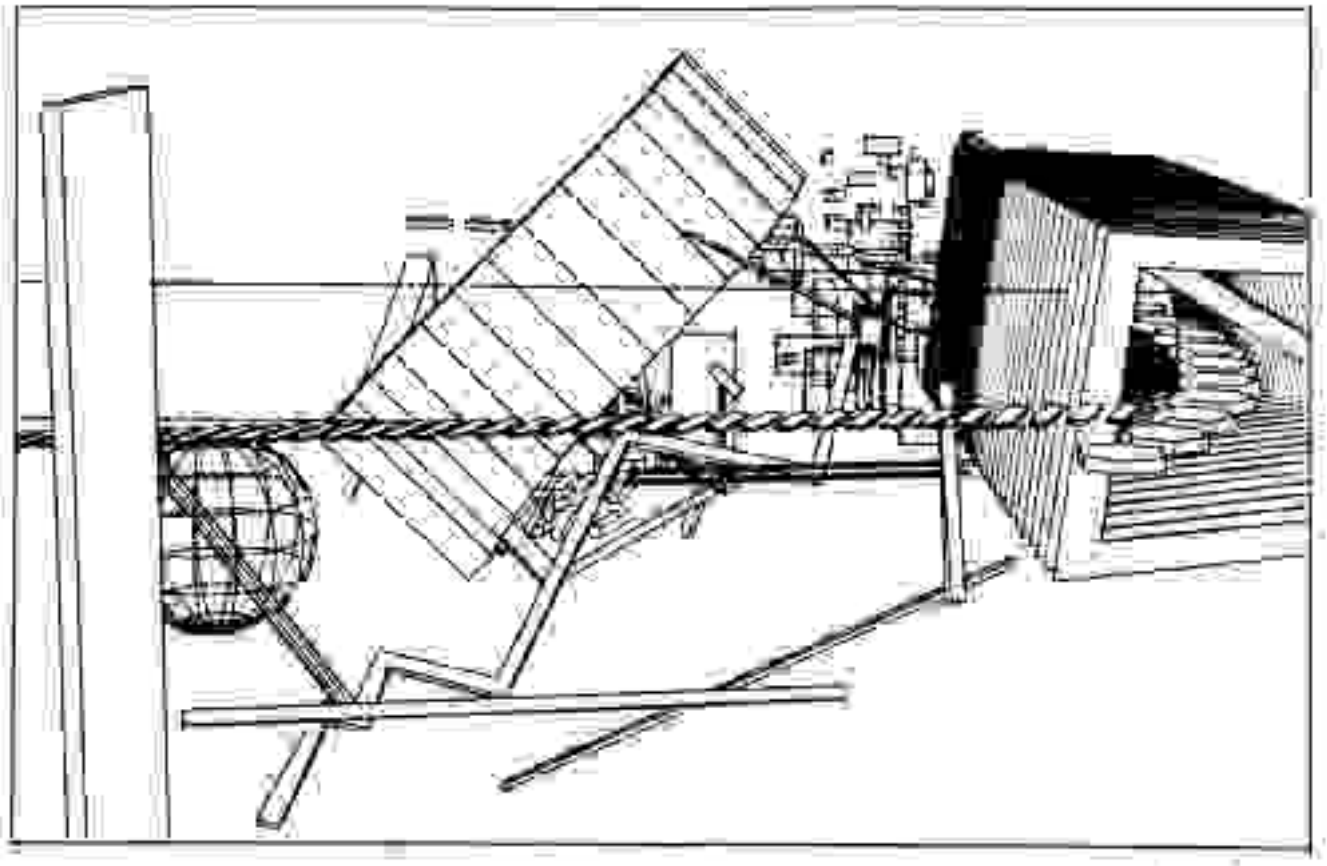




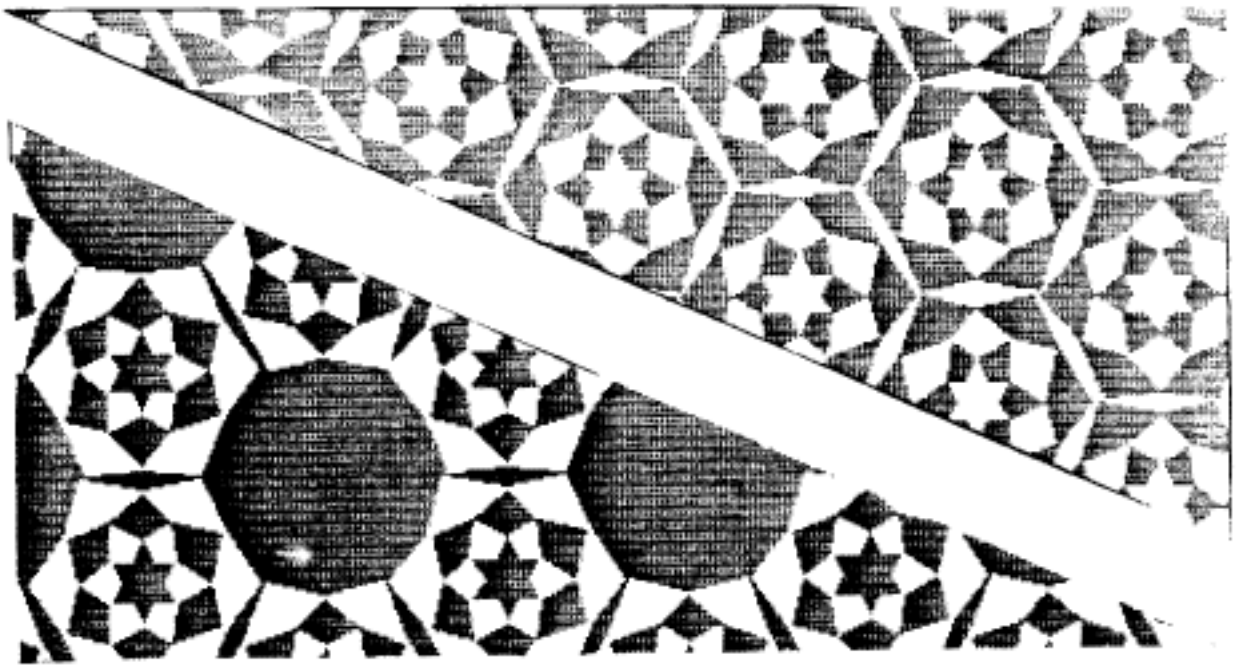
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